

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

QIAN QI

ABSTRACT. Microcanonical preparation is proved after exact particle-number coefficient extraction. A canonical trajectory-polymer expansion supplies the full number/continuous Hessian, a compact-frequency gap and a speed-dependent integrable high-frequency estimate for the coefficient law. A degree-one finite saddle and a relative local theorem for regular CLT-wide shells yield the source-dependent constrained pressure and covariance Schur complement.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: NUMBER-FIRST COEFFICIENT EXTRACTION AND THE PRIMITIVE SHELL

The empty grand-canonical sector is never bounded by an integrable continuous Fourier estimate. The exact target particle-number coefficient is extracted first. Only then is continuous Fourier inversion performed. Every remaining monomial contains exactly N_ε labels, which supplies the smoothing and removes the empty-sector obstruction.

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1.1. Grand-canonical and canonical generating functions. Let H denote the bounded particle-path and actual-contact source supplied by the grand-canonical part of B2. Let z be the activity and let Y^ε be the vector of continuous initial constraints (momentum, energy, and the declared macro statistics). Define

$$\mathcal{Z}_\varepsilon(z, H, u) = \mathbb{E}_\varepsilon^{\text{gc}} \left[z^{N^\varepsilon} e^{\mu_\varepsilon \langle H, X^\varepsilon \rangle + i \mu_\varepsilon u \cdot Y^\varepsilon} \right]$$

after dividing by the source-zero normalization. The exact-number partition function is

$$Z_{\varepsilon, N}(H, u) = [z^N] \mathcal{Z}_\varepsilon(z, H, u).$$

Theorem 1.1 (Number-coefficient trajectory-polymer expansion). *For $T \leq T_*$ and H in a fixed bounded real or complex source ball, $Z_{\varepsilon, N}(H, u)$ has a normally convergent canonical polymer expansion, uniform for N/μ_ε in a compact density interval. The logarithm divided by μ_ε converges locally uniformly, with all source and multiplier derivatives, to a canonical pressure $Q_N(H, u)$. The coefficient is obtained by a steepest contour through the unique activity saddle; no zero-particle or wrong-number sector appears in the continuous characteristic function.*

Proof. Expand the real-trajectory grand partition function into connected polymers and apply Cauchy's formula in z . Normal convergence on an annulus about the positive saddle permits coefficient extraction term by term. The saddle equation fixes exactly N labels, and the Gaussian activity direction gives the standard $N^{-1/2}$ coefficient. Tree-graph bounds and the B2 all-contact majorant are uniform on the compact density/source chart. Cauchy estimates then give convergence of every derivative. Since $[z^N]1 = 0$ for $N > 0$, the empty atom is removed before u is integrated. $\square$

1.2. The exact constraint Hessian and finite saddle. Let

$$C(x, v) = \left(1, v, \frac{|v|^2}{2}, \chi_1(x, v), \dots, \chi_m(x, v) \right).$$

Assume its affine span on the positive one-particle activity contains the full declared constraint space. For a multiplier λ write the exact finite pressure $Q_\varepsilon(H, \lambda)$.

Lemma 1.2 (Uniform finite-volume mean map). *On a common compact multiplier domain,*

$$cI \leq D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \leq CI.$$

The number coordinate and all cross terms are included. For every regular relative-interior target a there is a unique finite saddle $\lambda_{\varepsilon, H, a}$ satisfying $D_\lambda Q_\varepsilon(H, \lambda) = a$, and these saddles converge locally uniformly to $\lambda_{H, a}$.

Proof. The finite Hessian is the activity second moment

$$\frac{1}{\mu_\varepsilon} \text{Cov}_{H, \lambda} \left(\sum_i C(z_i), \sum_i C(z_i) \right),$$

not a normalized one-mark covariance. The canonical coefficient expansion contains a linear fraction of labels in finitely many phase-space patches on which the vectors C span the constraint space. The polymer correction is small on the short-time/source chart, giving the lower bound; Gaussian moments give the upper bound. Strict convexity and properness on the relative- interior chart yield the unique saddle by the degree-one mean map. Local uniform pressure convergence and the inverse-function theorem give saddle convergence. $\square$

1.3. Full continuous Fourier control after number extraction. Let $\phi_{\varepsilon,N,H}(u)$ be the continuous characteristic function under the exact-number, source-tilted canonical law centered at its finite saddle.

Theorem 1.3 (Integrable canonical Fourier theorem). *There are $\delta, R, c > 0$ and, for every $s > d + 3$, constants C_s such that uniformly on the regular source/target chart:*

(i) for $|u| \leq \delta$,

$$|\phi_{\varepsilon,N,H}(u)| \leq e^{-cN|u|^2};$$

(ii) for $\delta \leq |u| \leq R$,

$$|\phi_{\varepsilon,N,H}(u)| \leq e^{-cN};$$

(iii) for $|u| > R$,

$$|\phi_{\varepsilon,N,H}(u)| \leq C_s(1 + |u|)^{-s-cN_0} + C_s e^{-cN}(1 + |u|)^{-s},$$

where $N_0 \geq 1$ is fixed and the first exponent may be replaced by $s + cN$ on the good-label sector.

The right side in (iii) is integrable and its integral is $o(N^{-1/2})$. The same theorem holds jointly with the remaining lattice angles.

Proof. The small-frequency estimate is the finite Hessian expansion in Lemma 1.2. On a compact annulus, equality in the triangle inequality would force a nontrivial affine character of C to be constant on every spanning patch; the affine-span assumption excludes this, and the canonical polymer expansion preserves the gap.

For large u , examine each exact- N polymer monomial. A dependency-graph matching lemma gives one of two alternatives. Either a linear number of labels are leaves whose continuous marks can be integrated independently conditional on the remaining graph, producing a power $(1 + |u|)^{-s-cN}$; or the graph contains a linear number of extra polymer bonds, whose absolute activity is e^{-cN} . In the latter sector at least one label remains, and s integrations by parts in that label give the factor $(1 + |u|)^{-s}$. Thus even the exponentially small bad sector has frequency decay; no constant is integrated over an infinite Fourier line. Summing the normally convergent canonical polymers proves (iii). The lattice angles are compact and are handled by the same annular character argument. $\square$

1.4. The sharp primitive-shell coefficient. Let the particle number be the exact integer N_ε , and under the finite saddle write

$$\sqrt{\mu_\varepsilon}(Y^\varepsilon - a') \in W_\varepsilon,$$

where W_ε is a centered Lipschitz domain whose Gaussian boundary mass tends to zero and whose Gaussian mass $\gamma_{H,a}(W_\varepsilon)$ satisfies

$$\gamma_{H,a}(W_\varepsilon) \gg \mu_\varepsilon^{-K}$$

after comparison with the explicit Fourier remainder. Expanding centered boxes with $\sqrt{\mu} \delta_\varepsilon \rightarrow \infty$ are included.

Theorem 1.4 (Uniform mixed local coefficient). *Uniformly on compact regular source and target sets,*

$$\mathbb{P}_{H,\lambda_\varepsilon,H,a}(N = N_\varepsilon, \sqrt{\mu_\varepsilon}(Y - a') \in W_\varepsilon) = \mu_\varepsilon^{-1/2} c_{H,a} \gamma_{H,a}(W_\varepsilon) (1 + o(1)),$$

with $0 < c \leq c_{H,a} \leq C$. The error is relative to the displayed Gaussian mass.

Proof. Cauchy coefficient extraction in the number variable is performed first. On the canonical coefficient, Fourier inversion in the continuous coordinates has a Gaussian central region, an exponentially small compact annulus, and an integrable high-frequency tail by Theorem 1.3. Smooth inner and outer approximations of W_ε have boundary error controlled by its Gaussian boundary mass. The assumed comparison with the explicit remainder makes every noncentral term o of the main coefficient. The activity saddle contributes the one-dimensional lattice coefficient $c_{H,a} \mu^{-1/2}$. $\square$

1.5. Source-dependent microcanonical transfer.

Theorem 1.5 (Primitive-shell pressure and covariance). *Conditioning the grand-canonical hard-sphere law on the primitive shell gives*

$$Q^{\text{mc},a}(H) = Q(H, \lambda_{H,a}) - \lambda_{H,a} \cdot a - Q(0, \lambda_{0,a}) + \lambda_{0,a} \cdot a.$$

The initial empirical rate is the grand-canonical rate restricted to the constraint surface and normalized by its minimum. Its covariance is the Schur complement

$$Q_{HH} - Q_{H\lambda} Q_{\lambda\lambda}^{-1} Q_{\lambda H}$$

evaluated at $(0, \lambda_{0,a})$.

Proof. Apply the exact source-dependent change of measure separately to numerator and denominator. Theorem 1.4 shows that both shell coefficients are subexponential and gives their uniform ratio. The exponential parts are the two saddle values displayed above. Differentiating the saddle equation gives the Schur complement; Legendre duality gives the constrained initial rate. $\square$

Theorem 1.6 (Round-thirteen B1 closure). *The primitive hard-sphere shell has a source-dependent exact finite saddle, an integrable number-first Fourier coefficient theorem, a sharp regular-shell coefficient, and the constrained pressure/covariance of Theorem 1.5. The empty grand-canonical sector is*

eliminated by exact particle-number extraction rather than contradicted by a false pointwise bound.

Proof. Combine Theorems [1.1](#), [1.3](#), [1.4](#), and [1.5](#). □

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn